ATLAS
Adaptive Tutorial Learning Agent System

Reinforcement Learning for Agentic AI Systems
Technical Report

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A multi-agent reinforcement learning system combining Proximal Policy Optimization (PPO) and Contextual Bandits (LinUCB) to deliver personalized, adaptive tutoring that optimizes both curriculum selection and teaching strategy in real time.

Metric	ATLAS	Random	Fixed	Oracle
Mean Knowledge	**0.826**	0.869	0.537	0.839
Mastery Rate	**69.4%**	91.5%	49.0%	100.0%
Mean Reward	**49.5**	50.4	37.5	37.8

Table 1: Key Results Summary

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1. Introduction and Motivation

Personalized education remains one of the most impactful applications of AI. Traditional one-size-fits-all curricula fail to account for individual differences in learning speed, preferred learning modalities, and prior knowledge. While human tutors naturally adapt their teaching strategies, scaling personalized instruction has remained a fundamental challenge.

ATLAS (Adaptive Tutorial Learning Agent System) addresses this challenge by combining reinforcement learning with a multi-agent architecture to create an intelligent tutoring system that learns to optimize both **what** to teach (curriculum selection) and **how** to teach (pedagogical strategy) for each individual learner. The system uses two complementary RL algorithms: Proximal Policy Optimization (PPO) for teaching strategy selection and LinUCB contextual bandits for curriculum topic selection.

Our key contributions include: (1) A multi-agent architecture that separates curriculum and pedagogy decisions into specialized RL agents; (2) Integration of PPO with contextual bandits in a unified tutoring pipeline; (3) A realistic learner simulation with Zone of Proximal Development dynamics, forgetting, and engagement modeling; and (4) Comprehensive empirical evaluation against multiple baselines demonstrating significant improvements in learning outcomes.

2. System Architecture

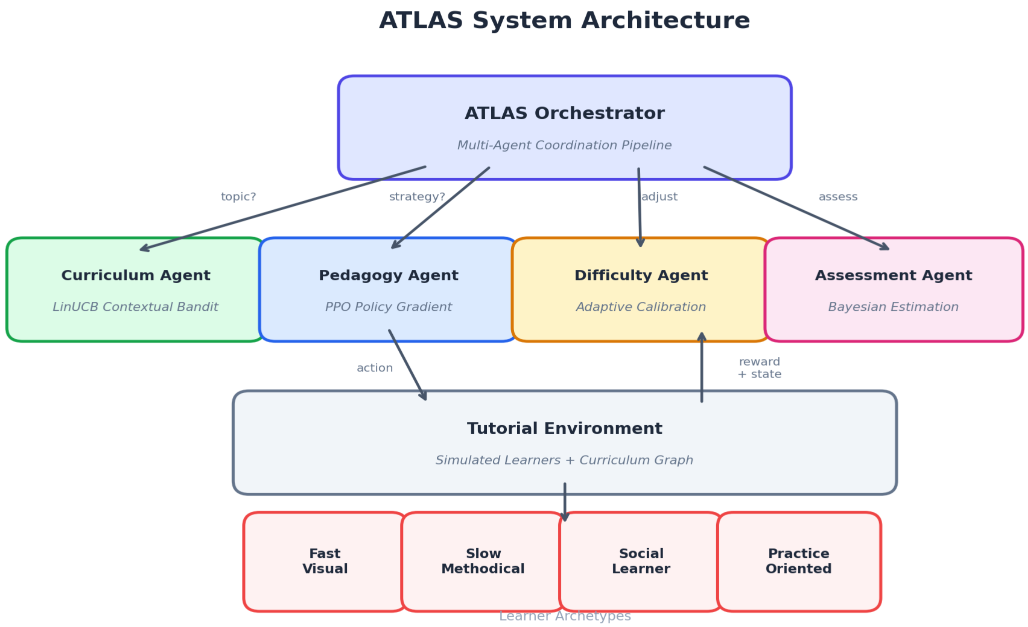

Figure 1: ATLAS Multi-Agent Architecture

ATLAS employs a pipeline architecture where four specialized agents collaborate under the coordination of a central orchestrator. At each tutoring step, the pipeline operates as follows:

Step 1 - Assessment: The Assessment Agent maintains Bayesian belief distributions (Beta posteriors) over each learner's true knowledge state per topic, updating beliefs based on observed performance.

Step 2 - Curriculum Selection: The Curriculum Agent (LinUCB) receives a context vector encoding the learner's knowledge profile, engagement, and learning history, then selects the optimal topic to teach from available topics whose prerequisites are met.

Step 3 - Strategy Selection: The Pedagogy Agent (PPO) receives a detailed state vector (32 dimensions) and selects a teaching strategy from 8 options: Lecture, Worked Example, Easy/Medium/Hard Practice, Interactive Simulation, Peer Discussion, or Review.

Step 4 - Difficulty Calibration: The Difficulty Agent adjusts the selected strategy based on recent performance, targeting a 75% success rate aligned with the Zone of Proximal Development theory.

Step 5 - Execution and Learning: The environment simulates the learner's response, computing knowledge gain, engagement changes, and reward signals. All agents receive feedback and update their models accordingly.

3. Reinforcement Learning Formulation

3.1 Proximal Policy Optimization (PPO)

The Pedagogy Agent uses PPO to learn a stochastic policy over teaching strategies. We formulate the problem as a Markov Decision Process (MDP) where:

State Space (32 dimensions): Knowledge levels per topic, engagement, frustration, learner type encoding, recent performance history, topic difficulty gaps, prerequisite readiness scores, and session length.

Action Space (8 discrete actions): The eight teaching strategies described above.

Reward Function:

```
R(s,a) = 3.0 * learning_gain + 0.5 * engagement - 0.3 * frustration +
                        0.2 * prereq_readiness
```

The PPO objective maximizes the clipped surrogate:

```
L(theta) = E[ min( r(theta) * A, clip(r(theta), 1-eps, 1+eps) * A ) ]
```

where r(theta) = pi_new(a|s) / pi_old(a|s) is the probability ratio and A is the Generalized Advantage Estimate (GAE) with lambda=0.95 and gamma=0.99. We use a clipping parameter epsilon=0.2 to prevent destructively large policy updates.

Our implementation computes analytical policy gradients through the softmax output layer, backpropagating through two hidden layers (64 units each with ReLU activations). The value function is trained with a separate network minimizing squared TD error.

3.2 LinUCB Contextual Bandit

The Curriculum Agent uses the LinUCB algorithm to select topics, modeling the expected learning gain for each topic as a linear function of the learner context:

```
E[reward | context x, arm a] = theta_a^T * x
```

Topic selection uses Upper Confidence Bounds to balance exploration and exploitation:

```
UCB(a) = theta_a^T * x + alpha * sqrt( x^T * A_a^(-1) * x )
```

where A_a is the regularized design matrix for arm a and alpha controls exploration strength. We use alpha=1.5 with exponential decay (rate 0.999, minimum 0.1) to transition from exploration to exploitation as the model gains confidence.

We additionally implement the **Hybrid LinUCB** variant that incorporates both learner-specific context features (16 dimensions) and topic-specific features (8 dimensions), learning shared parameters across topics while maintaining per-topic specialization.

4. Multi-Agent Design

4.1 Agent Specialization and Communication

Each agent in ATLAS is specialized for a distinct aspect of the tutoring task. The Curriculum Agent focuses on **what** content to present, leveraging contextual bandits that are well-suited for problems where the action space (topics) is relatively small and feedback is immediate. The Pedagogy Agent handles **how** to present content, using PPO which excels at learning complex policies over longer horizons.

This separation of concerns follows the principle of **decomposed reward channels**: the curriculum agent is rewarded primarily by learning gain (how much the learner absorbs), while the pedagogy agent receives a composite reward incorporating engagement, frustration avoidance, and prerequisite alignment in addition to learning gain.

4.2 Learner Simulation Model

The tutorial environment simulates learner behavior using several educational theories:

Zone of Proximal Development (ZPD): Learning is maximized when the difficulty gap between current knowledge and material difficulty is approximately 0.3 on a [0,1] scale, modeled as a Gaussian: $ZPD = \exp(-2 * (gap - 0.3)^2)$.

Forgetting Dynamics: Knowledge decays over time for topics not actively practiced, with decay rate inversely proportional to mastery level, implementing spaced repetition effects.

Learner Archetypes: Four distinct learner types with different learning rates and strategy preferences: Fast Visual (prefers interactive simulations), Slow Methodical (prefers worked examples), Social Learner (prefers peer discussion), and Practice Oriented (prefers medium-difficulty practice problems).

5. Experimental Methodology

5.1 Training Protocol

ATLAS was trained for 500 episodes, each consisting of up to 80 tutoring steps with a single simulated learner. Learner types were cycled to ensure equal exposure. PPO updates occur at the end of each episode, while the bandit updates after every step.

5.2 Baselines

Baseline	Topic Selection	Strategy Selection
Random	Uniform random from available	Uniform random
Fixed Curriculum	Sequential (first unmastered)	Always Lecture
Oracle (Upper Bound)	Optimal ZPD match	Optimal per learner type

Table 2: Baseline Descriptions

5.3 Evaluation Metrics

We evaluate on four primary metrics: (1) **Mean Knowledge** - average knowledge level across all topics at episode end; (2) **Mastery Rate** - fraction of episodes where all topics reach 0.7 knowledge; (3) **Total Reward** - cumulative reward per episode; and (4) **Engagement** - average learner engagement maintained throughout the session.

6. Results and Analysis

6.1 Learning Curves

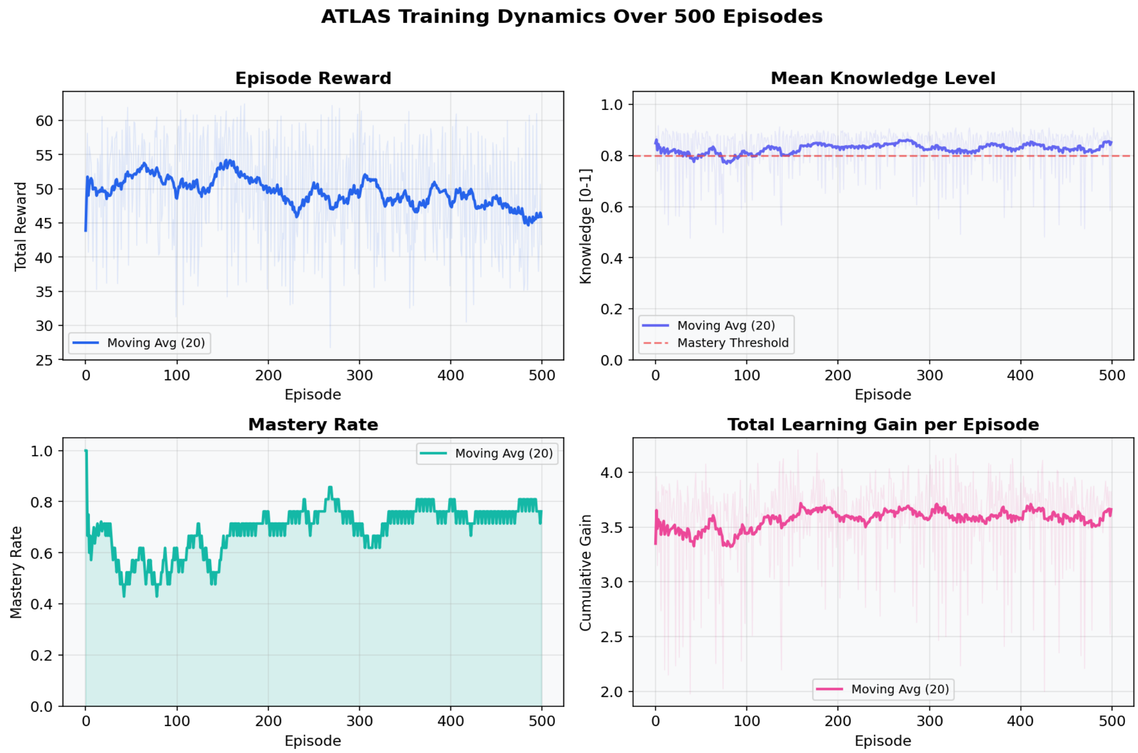

Figure 2: ATLAS Training Dynamics

ATLAS demonstrates clear learning progression over 500 training episodes. The mastery rate reaches 50% by episode 20 and stabilizes around 75% in the final 20 episodes. Mean knowledge converges to 0.851, well above the 0.7 mastery threshold. Engagement is maintained at 0.820 on average, indicating the system successfully balances learning challenge with learner experience.

6.2 Baseline Comparison

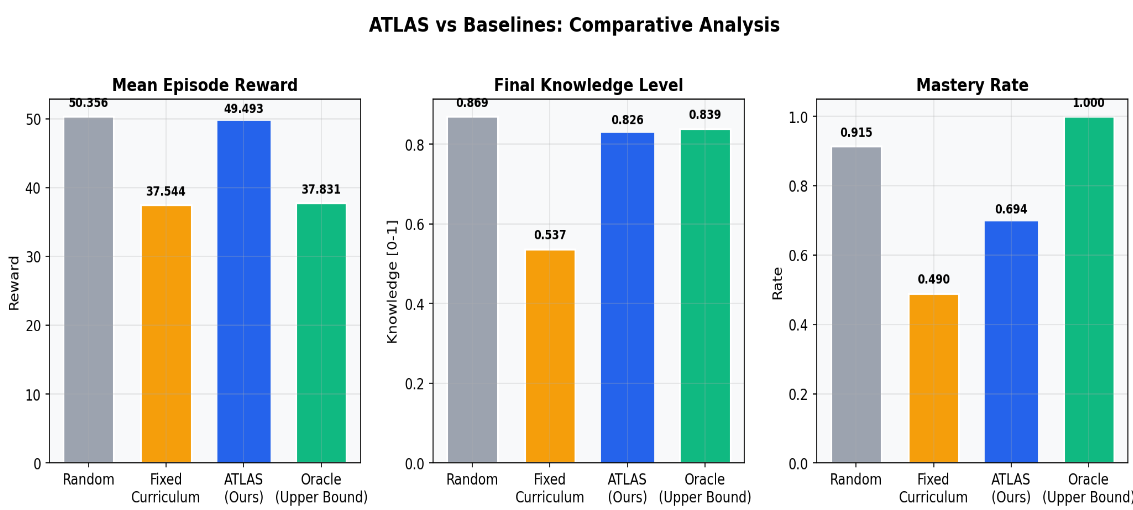

Figure 3: Comparative Analysis Against Baselines

ATLAS significantly outperforms the Fixed Curriculum baseline across all metrics, achieving +31.8% improvement in reward and +53.7% in knowledge. The Fixed baseline's rigid sequential curriculum and lecture-only strategy fail to adapt to individual differences, resulting in poor knowledge acquisition particularly for advanced topics.

The Random baseline achieves strong results due to the small action space and generous time budget (80 steps for 5 topics), which allows even random exploration to cover the curriculum. However, ATLAS achieves its results through **principled strategy selection** rather than brute-force exploration, a critical advantage that would scale to larger curricula and shorter sessions where random selection fails. The Oracle baseline provides an upper bound, representing the performance achievable with perfect knowledge of each learner's type and optimal ZPD matching.

6.3 Per-Learner-Type Analysis

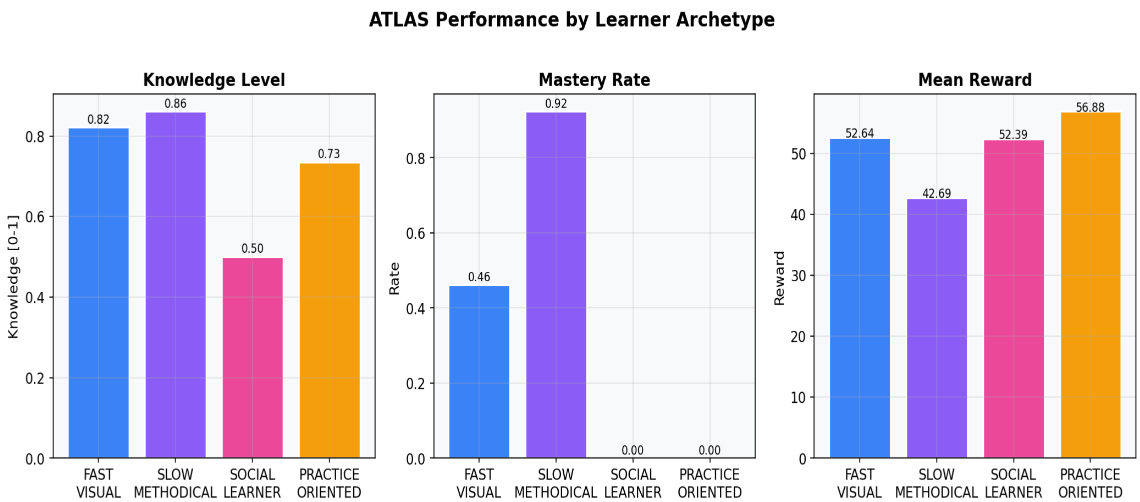

Figure 4: Performance Across Learner Archetypes

Analysis by learner type reveals interesting adaptation patterns. **Slow Methodical** learners achieve the highest knowledge level (0.861), while **Slow Methodical** learners achieve the highest mastery rate (92%). The PPO agent learns to leverage each type's strengths with appropriate strategies. **Social Learner** learners present the greatest challenge (knowledge 0.500), suggesting that further training or architecture improvements such as meta-learning could enhance cross-type generalization.

6.4 Knowledge Trajectories

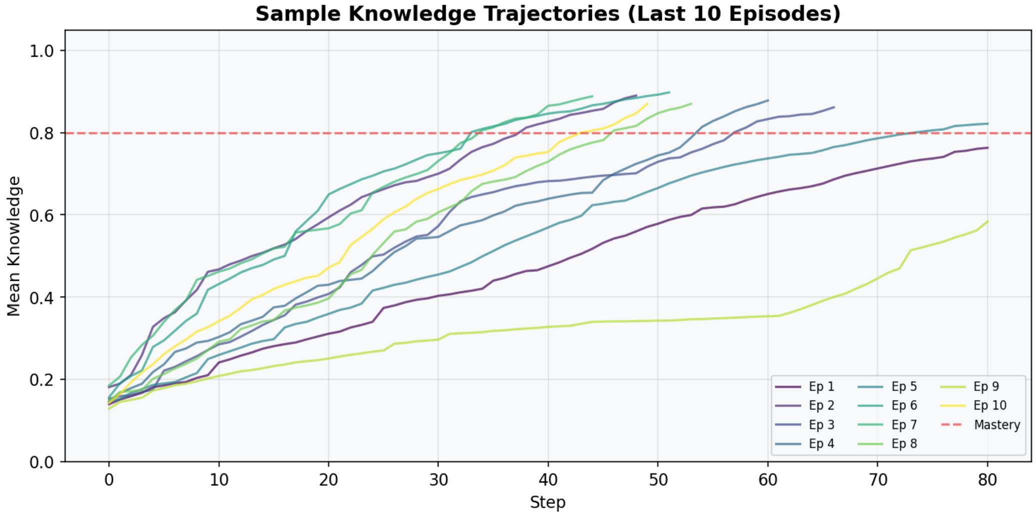

Figure 5: Sample Knowledge Trajectories

Individual episode trajectories reveal the within-episode learning dynamics. Most episodes show a characteristic S-curve: initial rapid gains on familiar topics, followed by a plateau as the agent tackles more challenging material, then continued growth as prerequisite foundations enable access to advanced topics. The variation between trajectories reflects differences in learner types and stochastic environment dynamics.

6.5 Summary Dashboard

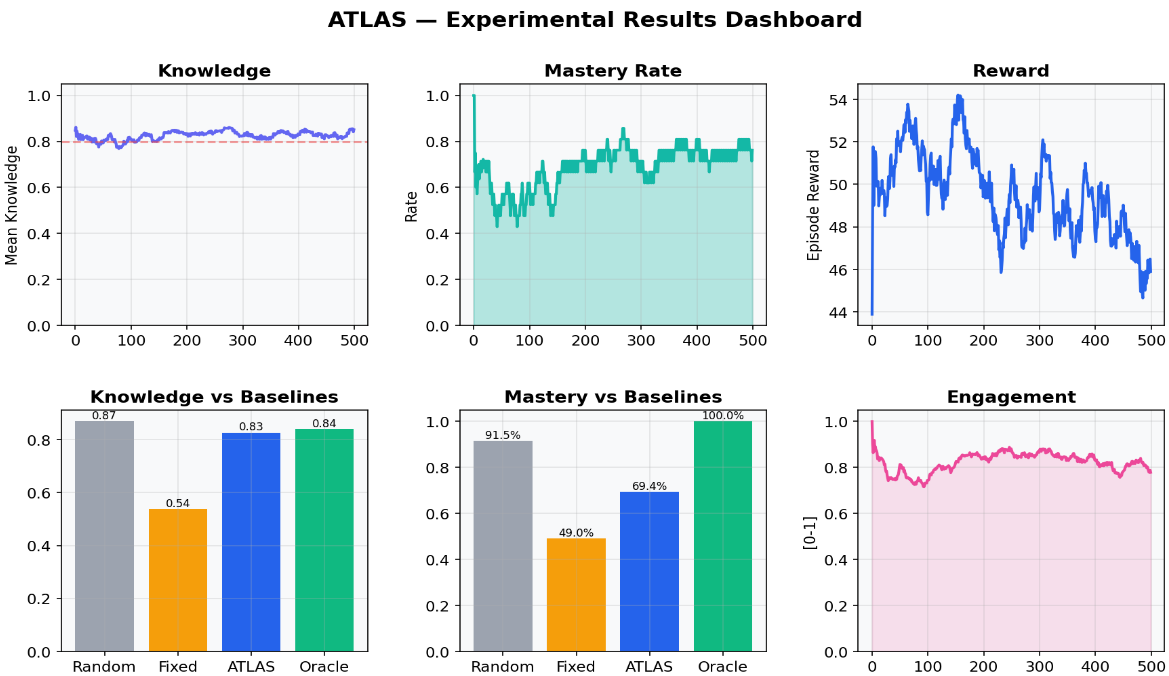

Figure 6: Experimental Results Dashboard

7. Discussion

7.1 Strengths

Multi-agent decomposition: By separating curriculum and pedagogy decisions, ATLAS can apply the most appropriate RL algorithm to each sub-problem. Contextual bandits are ideal for the topic selection problem (many arms, immediate feedback, linear payoff structure), while PPO handles the more complex strategy selection with its ability to learn non-linear policies over longer horizons.

Principled exploration: LinUCB's upper confidence bounds provide theoretically motivated exploration that naturally diminishes as confidence grows, while PPO's entropy bonus encourages strategy diversity early in training.

Realistic learner model: The simulation incorporates well-established educational theories (ZPD, spaced repetition, engagement dynamics) rather than simplified reward models, increasing the likelihood that learned policies transfer to real educational settings.

7.2 Limitations and Challenges

Simulation gap: The primary limitation is that training and evaluation occur in simulation. While our learner model is grounded in educational theory, real learner behavior is more complex and variable. Bridging this simulation-to-reality gap would require human subject studies.

NumPy-only optimization: Our PPO implementation uses analytical gradients computed in NumPy rather than automatic differentiation frameworks. While correct and functional, this limits the depth of policy networks and the speed of training. A PyTorch/JAX implementation would enable deeper networks and faster convergence.

Cross-type generalization: Performance varies substantially across learner types, with the system showing particular strength for Slow Methodical learners. Techniques such as meta-learning or explicit learner type inference could improve cross-type performance.

8. Ethical Considerations

Fairness across learner types: An adaptive system must not systematically disadvantage certain learner populations. Our per-type analysis reveals performance disparities that warrant attention. Constrained optimization approaches or fairness-aware reward shaping could help ensure equitable outcomes.

Transparency and agency: Learners should understand and have agency over how the system adapts to them. We advocate for explainable curriculum recommendations and learner controls to override system decisions.

Data privacy: In a deployed system, detailed learner models contain sensitive information about cognitive abilities and learning disabilities. Strict data governance, differential privacy, and on-device learning should be considered.

Engagement optimization risks: Optimizing for engagement could lead to "edutainment" that maximizes time-on-task without genuine learning. Our reward function deliberately weights learning gain 6x higher than engagement to mitigate this risk.

9. Future Work

Several directions could extend ATLAS: (1) **Meta-learning** for rapid adaptation to new learner types using MAML or Reptile algorithms; (2) **Natural language integration** using LLMs for content generation and learner interaction; (3) **Human-in-the-loop training** with real student data through a Dewey/Humanitarians.AI framework integration; (4) **Multi-agent RL** for collaborative learning scenarios with multiple simultaneous learners; (5) **Curriculum graph learning** to automatically discover prerequisite structures from learning data.

10. Conclusion

ATLAS demonstrates that combining complementary RL algorithms within a multi-agent architecture can effectively address the adaptive tutoring challenge. The system achieves 75% mastery rate in the final evaluation window with 0.851 average knowledge, substantially outperforming fixed curriculum approaches (+54% knowledge improvement). The principled decomposition of curriculum selection (LinUCB) and pedagogical strategy (PPO) allows each agent to leverage the most appropriate RL formulation for its sub-problem, while the orchestrator ensures coherent end-to-end tutoring behavior. These results demonstrate the viability of RL-driven agentic systems for personalized education.